

Personal Statement 1

I am currently a student on the UCL/Birkbeck MSc in Cognitive Neuropsychology and Neuroscience. Prior to this I gained a First Class BSc in Psychology at UCL, and was included in the Dean's List for outstanding academic performance. I am applying to do a PhD at the Social, Genetic and Developmental Psychiatry centre because I am keen to explore genetic influences on impaired cognitive function in developmental disorders such as ADHD. The discovery of genes linked to symptoms fascinates me because it holds potential for illuminating the etiological basis of psychiatric disorders, as well as improving diagnostic specificity. Studying at the SGDP would give me the opportunity to work with the foremost experts in the field of behavioural and molecular genetics; and provide excellent training in experimental techniques.

Over the course of my MSc I have become interested in research identifying the neural basis of individual differences in executive function, and developmental disorders such as ADHD. I am keen to use endophenotype measures associated with genetic risk, such as response inhibition or variability in response latencies (Kuntsi et al., 2001), to facilitate the discovery of genes linked to ADHD. I would also like to parse these larger performance phenotypes into component cognitive processes; such as modulation of behaviour based on dynamic goals and inhibition of habitual responses (Aron & Poldrack, 2005; Green et al., 2008). This may further improve the power to identify genetic effects on aspects of performance.

Another aspect of ADHD which I would like to study is impulsivity; the degree to which immediate rewards are weighted relative to longer term rewards or punishments. I am keen to explore differential weighting using behavioural experiments on genotyped samples; as well as pharmacological interventions to measure differential sensitivity to neuro-chemical levels in patients versus control subjects. Genetics evidence suggests that multiple catecholaminergic systems may be dysfunctional in ADHD; as such pharmacological manipulations could shed light on these factors.

At present, I am working with Dr Jonathan Roiser and Dr Essi Viding on my MSc project: an investigation of how brain function differs in healthy individuals who possess a common genetic polymorphism linked to depression, the serotonin transporter linked polymorphic region (5-HTTLPR) (Heils et al., 1996, Schinka et al., 2004). The aim of this project is to understand the genetic susceptibility mechanism which alters individuals' reactivity to negative stimuli and may as a result contribute to risk for affective disturbances. Such findings can be used to promote 'environmental engineering' techniques to offset genetic risk; and develop novel treatment targets for pharmacotherapy.

Preliminary results from my MSc project suggest that risk allele carriers show differential responsiveness in the sub-genual anterior cingulate; a region which is hyper-active in depressed patients (Drevets et al., 1997); and is thought to have a key modulatory role in emotional reactivity. Shortly I will begin using an effective connectivity based approach called Dynamic Causal Modelling to assess how activity in this region modulates coupling between aspects of the emotional processing network.

To conclude, studying at the Social Genetic and Developmental Psychiatry Centre would provide me with a fantastic opportunity to work in an inter-disciplinary environment. It would be the ideal setting to develop my skills as a scientist, further my academic interests, and subsequently allow me to pursue a career in academia.